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Commentary

Rethinking structural safety for a volatile century

Gang Xu^a, Tong Guo^{a,*}, Ai-Qun Li

Key Laboratory of Concrete and Prestressed Concrete Structures of the Ministry of Education, Southeast University, Nanjing 211189, China



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ABSTRACT

Most structural design frameworks remain rooted in outdated assumptions: that hazards occur in isolation, materials do not degrade, and failure can be prevented through overdesign. In an era marked by accelerating climate volatility, cascading disasters, and mounting carbon constraints, such logic is increasingly untenable. From earthquake–flood sequences to heat-induced corrosion and permafrost collapse, modern infrastructure faces compound and evolving risks that current codes fail to anticipate. At the same time, conventional approaches often achieve safety through carbon-intensive construction, exacerbating the very environmental crises they must endure. This rethinking aligns with emerging paradigms in resilience engineering, disaster risk reduction, and adaptive infrastructure design, which similarly emphasize systems thinking, robustness, and long-term functionality under uncertainty. We argue for a fundamental redefinition of structural safety—one that prioritizes resilience over resistance, adaptability over rigidity, and sustainability over excess. This requires innovative tools (AI-driven diagnostics, lifecycle modeling frameworks), new materials (durable, repairable, low-carbon-footprint systems), and quantifiable metrics (recovery duration, embodied carbon emissions, residual functionality assessment). Structural engineers must now lead as strategists of climate-ready, socially responsive infrastructure. The goal of the next generation of design is not merely to ensure that structures survive, but that they serve, recover, and regenerate in a volatile century.

1. Time to rethink safety

For much of the past century, structural safety has been defined by a narrow objective: preventing collapse during a rare, isolated event such as a major earthquake or hurricane. This has shaped a global design philosophy centered on the ultimate limit state—ensuring that structures maintain sufficient strength and stiffness under a single, extreme hazard. While this approach significantly improved safety in the 20th century, it now falls short of meeting the challenges posed by today's climate and risk landscape.

Recent disasters underscore the limitations of this paradigm. The 2011 Great East Japan Earthquake triggered not only catastrophic ground shaking but also tsunamis, nuclear failures, and widespread energy disruptions—a convergence of hazards that exceeded the assumptions of traditional design models. Likewise, the 2019 Venice floods, driven by a combination of sea level rise and intense rainfall, overwhelmed urban infrastructure designed for simpler, historical conditions. These cases are no longer exceptions but increasingly the norm.

Compounding this, the reliance on steel- and concrete-intensive solutions—the default in modern construction—has contributed substantially to global carbon emissions [1]. Structures built to withstand extreme scenarios are often overdesigned for their day-to-day functions, locking in high embodied carbon while offering limited adaptability or

functional resilience. The Burj Khalifa, for example, demonstrates both engineering prowess and the carbon cost of scale and redundancy.

A shift in mindset is needed. Structural design must now move beyond isolated resistance toward integrated resilience. Buildings and infrastructure must be conceived as adaptive systems—capable of withstanding multi-hazard conditions, supporting rapid functional recovery, and minimizing environmental impact. Milan's Bosco Verticale offers one glimpse of this future: a high-rise designed not only for wind and seismic loads, but also for urban heat mitigation and air quality improvement through integrated vegetation.

This redefinition of structural safety requires us to rethink performance goals, design tools, and regulatory frameworks. The 21st century demands infrastructure that is not only strong, but regenerative—prepared to withstand disruption, serve communities, and reduce its own contribution to planetary risk.

2. Design philosophy built on blind spots

Despite incremental progress, the foundation of most structural codes remains largely unchanged: they assume materials are constant, hazards are independent, and damage is binary—either avoided or catastrophic. This narrow framing ignores the nuanced, often cumula-

* Corresponding author.

E-mail address: guotong@seu.edu.cn (T. Guo).

tive nature of risk. It also underestimates the central challenge of our time: compound hazards.

Here, we distinguish “residual functionality”—the level of usable service remaining immediately after a hazard—from “functional recovery,” which denotes the process and timeframe required to return to full or acceptable performance.

Disasters rarely occur in isolation. A building that survives an earthquake may be damaged beyond repair when a fire breaks out hours later. Foundations may be softened by floodwaters just days after a heatwave-induced drought. These were not hypothetical scenarios in the 2011 Great East Japan Earthquake or the 1994 Northridge Earthquake in California—both of which revealed the limits of single-hazard thinking.

Moreover, modern buildings face increasingly “silent degradations.” Climate-induced corrosion, thermal fatigue, and freeze–thaw damage all chip away at strength long before disaster strikes. Few codes mandate consideration of these effects. In effect, many structures today are evaluated as if new, with their maximum strength and minimum vulnerability—a dangerously optimistic assumption in a warming world.

This approach is increasingly insufficient given today’s compound risk context. Safety, as currently defined, overemphasizes resisting collapse under ideal conditions while neglecting recovery under real ones. It prizes resistance, yet disregards resilience. These shortcomings are not just theoretical—they have real-world consequences that are increasingly visible in post-disaster failures.

3. Hazards as the new normal

Climate volatility is no longer a distant threat; it is the defining background condition for today’s infrastructure. From record-breaking heatwaves and shifting precipitation patterns to thawing permafrost and coastal flooding, environmental stressors are acting with greater frequency, intensity, and simultaneity [2]. Structural design must move from treating volatility as an exception to treating it as the baseline.

The 2021 Henan floods in China followed a prolonged heatwave that had already weakened concrete and underground systems through microcracking and material fatigue. When the rain came—over 600 mm in a day—urban drainage systems failed, subway tunnels flooded, and buildings collapsed. The event was not exceptional in nature, but in sequence.

Similar cascading failures occurred in Hokkaido, Japan, in 2018, when an earthquake was swiftly followed by Typhoon Jebi. Landslides triggered on slopes already destabilized by shaking. Recovery efforts from one hazard were precluded by the onset of another.

In cold regions, freeze–thaw cycles are now more frequent and erratic. Structures in Northern Europe and Canada are seeing reduced service lives from accelerated fatigue, especially when de-icing salts increase corrosive exposure. Meanwhile, permafrost thaw in Siberia and Alaska is causing foundation failure in buildings, pipelines, and tanks, as in the 2020 Norilsk diesel spill, where collapsing supports released over 20,000 tons of fuel.

Yet structural models and codes still assume static baselines: soils don’t shift, materials don’t degrade, hazards don’t interact. That fiction is becoming more costly each year.

4. Toward structures that adapt, not just endure

The future of structural design lies in a fundamental shift from a narrow focus on strength and stability toward a more integrated paradigm—one that unites resilience, low-carbon performance, and climate adaptability. This transformation requires a holistic, multi-dimensional framework, reimagining structures not only as objects that resist damage, but as systems capable of absorbing shocks, facilitating recovery, and minimizing environmental impact across their full life-

cycle—e.g., by reducing embodied carbon in foundations and enabling material reuse at end-of-life.

First, multi-hazard resilience must become a foundational principle. Structures should not merely aim to avoid collapse under a single extreme event; instead, they must sustain critical functionality and allow for rapid recovery amid compound or sequential hazards [3]. For example, a building in a coastal seismic region might need to endure ground shaking, resist tsunami-induced scour, and remain functional through extreme heat or power outages. In such cases, robustness alone is insufficient—design must allow for controlled damage, targeted energy dissipation, and modular repairs, enabling continuity of use and swift post-disaster rehabilitation.

Second, structural safety must be evaluated through a lifecycle lens. Performance should be assessed not just at the moment of peak hazard, but across construction, operation, maintenance, hazard events, and end-of-life stages [4]. This approach encourages optimization over maximization, balancing structural demands with material efficiency, embodied carbon, operational emissions, and long-term reparability. For instance, instead of over-reinforcing a shear wall, a designer might choose a thinner wall integrated with replaceable energy-dissipating devices, achieving similar performance with less material and greater flexibility.

Third, the integration of climate-adaptive materials and systems is essential [5]. Traditional materials degrade faster under fluctuating environmental stressors such as temperature swings, humidity, salinity, and pollution. In contrast, durable concretes with low permeability, fiber-reinforced polymers, self-healing coatings, and self-centering connections can dramatically extend structural lifespan and enhance post-hazard recovery. New joint technologies—such as sliding bearings or replaceable fuses—can isolate damage, reduce residual drifts, and simplify post-event inspection and repair.

Finally, the use of AI-driven and data-informed tools holds transformative promise [6]. Machine learning models trained on historical damage records and environmental data can help predict deterioration, optimize material allocation, and simulate structural responses under probabilistic multi-hazard scenarios. When coupled with digital twins and sensor networks, these systems can provide near-real-time diagnostics and guide adaptive interventions. However, challenges remain in the form of data scarcity, limited long-term field validation, and integration complexity across platforms, which must be addressed to realize the full potential of these technologies.

While the focus here is on forward-looking design, these principles also inform strategies for retrofitting and adaptive reuse of existing infrastructure, offering pathways to enhance resilience and reduce emissions in the built environment. In this emerging paradigm, damage is not synonymous with failure, but a designed and manageable phase within the lifecycle. Structural safety, redefined, is not the absence of impact, but the capacity to resist, adapt, and regenerate in the face of an uncertain and evolving world.

5. Time to update standards, databases, and mindsets

Achieving climate-adaptive, low-carbon structural resilience is no longer a question of isolated innovation—it is a systemic imperative. While engineering tools and materials are rapidly advancing, the institutional and regulatory frameworks that govern structural design have largely remained static. Most existing codes are still grounded in deterministic design philosophies, focused on peak load resistance under idealized, singular events. They seldom account for compound hazard sequences, gradual climate-driven degradation, or post-event functional recovery. To respond to today’s risks, new metrics and evaluation criteria must be formally embedded into standards and design protocols, including recovery time, residual functionality, reparability, and embodied carbon performance.

Beyond codes, the lack of integrated, high-quality multi-hazard performance databases poses a major barrier. Current datasets are frag-

mented across disciplines and typically focus on either acute events or long-term deterioration, rarely both. A robust transition demands new interdisciplinary databases that capture the full lifecycle of structural behavior, including environmental degradation mechanisms (e.g., corrosion, freeze-thaw, thermal fatigue), disaster sequences (e.g., earthquake-typhoon-heatwave cascades), and carbon emissions at each stage from construction through demolition. These databases would support AI-driven modeling, improve probabilistic risk assessments, and enable performance-based benchmarking aligned with future climate realities.

Meeting these challenges requires broad, cross-sector collaboration that integrates insights from multiple fields—including climate science, material engineering, data science, and public policy—to align technical innovation with environmental and societal needs. Public and private sectors must work together to implement pilot projects, establish green infrastructure innovation hubs, and reform procurement standards to reward adaptability and carbon-conscious design, not just compliance with outdated codes.

Governments and regulatory bodies should take a leading role, by offering incentives for net-zero structural systems, mandating carbon and resilience reporting in approvals, and funding research into next-generation design standards. Educational institutions, meanwhile, must update curricula to equip future engineers with the tools to design for uncertainty, recovery, and sustainability.

Ultimately, the goal is not just to make structures safer for today's hazards, but future-ready for a century defined by volatility. Structural resilience must be redefined as the ability to endure, adapt, and regenerate, while minimizing environmental impact and supporting societal continuity in the face of an uncertain future.

6. The evolving role of structural engineers

In a century shaped by climate disruption, resource scarcity, and cascading hazards, structural safety can no longer be confined to the binary question of collapse under a rare event. It must evolve to reflect more complex objectives: maintaining critical function, enabling rapid recovery, minimizing environmental cost, and adapting to uncertain futures. This broader mandate redefines not only what structures must do, but also who structural engineers must become.

The structural engineer of today must transcend traditional disciplinary boundaries [7]. No longer just a seismic expert or code interpreter, they are called to be resilience strategists, climate risk translators, and systems integrators. They must navigate a landscape where hazards are multi-dimensional, materials must perform under stress and scrutiny, and design decisions have long-term carbon consequences. Their toolkit must now include climate projections, lifecycle modeling, AI-enabled design algorithms, and knowledge of circular construction. Their role extends from structural analysis to urban continuity planning, from finite element modeling to policy advocacy.

This expanded role is not just technical—it is ethical. The profession must adopt a new design ethos: one that values performance over prescription, flexibility over excess, and resilience over redundancy. It means designing not only to prevent collapse, but to support lives and livelihoods in the aftermath. It means understanding that failure is not always physical — sometimes it is systemic, and sometimes, it is failing to anticipate change. And the engineers who design them must be equipped to anticipate, adapt, and lead.

To meet this moment, engineering education, professional standards, and institutional missions must adapt. Future engineers must be trained not just in mechanics but in climate science, data analytics, stakeholder communication, and sustainable design thinking. Codes and contracts must reward recovery capacity and carbon efficiency, not just compliance. And the global engineering community must come together—not only to share technical knowledge, but to define and defend a shared vision for a livable future.

Ultimately, our structures will not be judged solely by their height or longevity, but by how lightly they burden the planet, how intelligently they respond to stress, and how quickly they restore the functions that communities depend upon [8]. This is the new frontier of structural design. And it begins with reimagining the engineer.

7. A call to action

In a world defined by volatility, structures that cannot adapt will not endure. And a design philosophy that fails to change will not protect us. Rethinking structural safety is not a matter of adding complexity—it is about aligning design with reality. We need a new infrastructure ethic: one rooted in resilience, sustainability, and justice. Our buildings should not just stand—they should serve, shelter, and regenerate. And the engineers who design them must be equipped to anticipate, adapt, and lead. The choice is not whether change is coming. It is whether we shape it or let it collapse upon us.

CRedit authorship contribution statement

Gang Xu: Writing – original draft. **Tong Guo:** Supervision. **Ai-Qun Li:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no conflicts of interest in this work.

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Author profile

Gang Xu (BRID: 08035.00.33791) is an associate professor of School of Civil Engineering at Southeast University. He received his Ph.D. in Civil Engineering from Southeast University in 2020 and joined the faculty in 2021. His research focuses on disaster prevention and mitigation technologies in civil engineering, addressing national needs for disaster resilience and resilient urban development.

Tong Guo (BRID: 09327.00.97017) is the Chair Professor and Dean of School of Civil Engineering at Southeast University. He was funded by National Science Fund for Distinguished Young Scholars in 2021. He works in multi-disciplinary research areas across civil-, structural-, mechanical-, architectural- engineering and computational mechanics with the specific research interests in the development and application of structural condition assessment and upgrade methods to civil structures, including: structural health monitoring, novel aseismic structural systems, life-cycle assessments of buildings and bridges and structural vibration control, etc.